Name: _____

# YEAR 11 SPECIALIST MATHS INVESTIGATIVE TASK 1

# Vectors

**Prerequisites**

- Basic understanding of vectors

1 cm or 0.5 cm graph paper is required. CAS calculators would be useful but are not essential.

**[Total: 70 marks]**

Exact answers should be given where possible, unless otherwise instructed. Surds should be simplified, including, where appropriate, being expressed with a rational denominator.

**1** $\mathbf{p}$ is the vector $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$ and $\mathbf{q}$ is a vector of magnitude 2 in the positive $x$ direction.

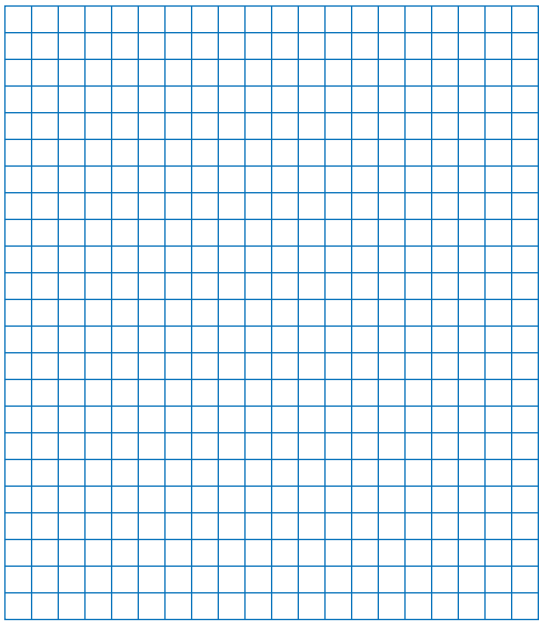
Use graph paper to draw vector diagrams for:

**a** $\mathbf{p}$ **b** $\mathbf{q}$ **c** $\mathbf{p} + \mathbf{q}$ **d** $\mathbf{p} - \mathbf{q}$

**e** $\hat{\mathbf{q}}$ **f** $2\mathbf{p} + 3\mathbf{q}$ **g** the projection of $\mathbf{q}$ on $\mathbf{p}$

**[7 marks]**

Graph paper for question 1



**2** Given that $\mathbf{a} = 2\mathbf{i} + \mathbf{j}$, $\mathbf{b} = 3\mathbf{i} - 4\mathbf{j}$ and $\mathbf{c} = \mathbf{i} + \mathbf{j}$, find

**a** $\mathbf{a} + \mathbf{b}$ **b** $\mathbf{a} - \mathbf{b}$ **c** $4\mathbf{b}$ **d** $|\mathbf{b}|$

**e** $\hat{\mathbf{b}}$ **f** $\mathbf{a} - 3\mathbf{b} + 2\mathbf{c}$ **g** $\mathbf{a} \cdot \mathbf{c}$

**h** the angle between **a** and **c** to the nearest degree

**i** the projection of **a** on **b**

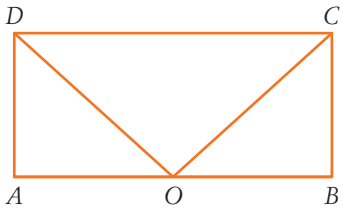
**j** orthogonal components of a parallel to and perpendicular to **c**

**k** a vector of magnitude 4 parallel to **c**

**l** a vector of magnitude 10 perpendicular to **b**

**[1 + 1 + 1 + 1 + 1 + 1 + 1 + 2 + 2 + 3 + 2 + 8 = 24 marks]**

**3** *ABCD* is a rectangle and *O* is the midpoint of *AB*.




Express each of the following in terms of **OC** and **OD**, showing all steps of reasoning.

**a** **CD**

**b** **OA**

**c** **AD**

**[2 + 3 + 4 = 9 marks]**

**4** The Keystone Cops have decided a heavy crate must be moved. They are rather disorganised, so they all push on it from different directions. Albert pushes to the north with a force of 450 N, Barry pushes to the east with 400 N and Charlie pushes to the north-west with 600 N. Donald notices it is rather crowded where these three are, so he goes round the other side and pushes south with a force of 500 N. A minimum force of 350 N is required to overcome friction and move the crate. Use CAS calculators where applicable to get exact answers and where necessary evaluate these approximately.

**a** Draw a diagram showing the forces acting on the box as a result of their efforts. Do not include any other forces like friction or gravity.

**b** What is the combined effect of Albert's and Donald's efforts?

**c** Draw a new diagram where Albert and Donald's forces are replaced by this combined effect.

**d** Use vector addition to find the (exact) magnitude of the overall effect of all their efforts? A vector addition diagram should be included.

**e** Will the crate move and if so, in what direction?

**[2 + 1 + 2 + 9 + 5 = 19 marks]**

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**5** One day Superman uses his telescopic vision to spy a criminal at a point $-3\mathbf{i} + \mathbf{j}$ from his starting point $O$ (0, 0) at the *Daily Planet*. He immediately flies toward the criminal at a speed of 100 km/h (relative to the atmosphere). At the same time a north-east wind is blowing at 30 km/h.

*Note*: a north-east wind is blowing *from* the north-east.

Let **i** represent 1 km in the easterly direction and **j** represent 1 km in the northerly direction.

**a** How far away is the criminal from Superman, and in what direction, to the nearest degree, should Superman fly to catch the criminal?

**b** Allowing for the wind, in which direction should Superman aim his flight so that he is flying directly towards the criminal?

**c** How long will it take Superman to get to the crook?

**d** As soon a Superman nabs the crook, Jimmy Olsen buzzes him to tell him that his arch enemy Lex Luther is holed up at $2\mathbf{i} + 5\mathbf{j}$. Express his displacement relative to Superman in terms of **i** and **j**.

**e** By the time Superman deals with Luther and returns to the *Daily Planet*, what is the total distance he has travelled?

**[5 + 9 + 5 + 1 + 2 = 21 marks]**

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